1. Business Problem, Objectives & Business Questions

Project: Retail Business Performance & Profitability Analytics **Company (fictional scenario, synthetic data):** RetailHub — a multi-city Indian retail chain 20 stores + online, 8 product categories, selling in 2024–2025 **Analyst:** Teja S **Dataset:** Synthetic retail dataset generated in Python (seed = 42) for this project — 10,000 orders across 20 stores and 8 categories, 2024–2025, modelled on a realistic Indian retail chain scenario. All results are calculated from this dataset.

1.1 The real-world business problem

RetailHub is growing in sales, but **growth is not turning into healthy profit**.

- Sales rose only modestly from 2024 to 2025: ₹5.60 Cr → ₹5.77 Cr (+2.9%)
- Overall profit margin is 17.52% — but that average hides the problem
- 1,483 of 10,000 order lines (14.8%) were sold at a **loss**, losing ₹13.20 lakh in total
- Festive months (Oct–Dec) bring the highest sales of the year, yet some of the **lowest margins** — Nov 2025: 12.78%, Nov 2024: 13.03%, Oct 2024: 14.45%, compared with 21.21% in Mar 2025
- Heavy discounting is the main suspect: lines discounted 20–30% earned a margin of just 0.64%, and lines discounted above 30% lost **-15.97%**

In short: **the business does not know where profit is actually made, where it leaks, and which levers — discount, category, store, returns — managers should pull first.**

Without this analysis, decisions stay guess-based: more discounts to chase sales, equal attention to all stores and categories, and no action on preventable returns.

1.2 Problem statement

> RetailHub's sales are growing slowly while profitability is being eroded by deep discounting, uneven category and store performance, and preventable returns. This project analyses 10,000 transactions from 2024–2025 across Excel, SQL, Python and Power BI to identify exactly where profit is created and lost, and recommends specific actions to improve margin without killing sales volume.

1.3 Why are we doing this project? (Business objectives)

#	Objective	How success is measured
1	Find the true drivers of profit and loss	Profit / margin broken down by category, product, store, segment, month
2	Stop discount-led profit leakage	Margin by discount band; a clear discount ceiling recommendation
3	Fix festive-season profitability	Oct–Dec sales vs margin comparison, year on year
4	Improve underperforming stores & categories	Store / category ranking on sales *and* margin, not sales alone
5	Reduce preventable returns	Return rate by category and reason; share of returns that are operational
6	Know the best customers	Sales, margin and repeat behaviour by customer segment
7	Give managers a repeatable view	Excel workbook + Power BI dashboard they can refresh monthly

1.4 How data analysis solves it

1. **Clean and validate first** (Excel / Power Query + Python checks) — so every number that follows can be trusted. Formula, FK, duplicate and missing-value checks all passed (see `data/validation_report.txt`).
2. **Ask the data structured questions** (SQL) — reproducible queries, not screenshots.
3. **Explore patterns humans miss** (Python EDA) — discount vs margin correlation is **-0.87**, a strong relationship no PivotTable alone would quantify.
4. **Make it usable for decision-makers** (Power BI) — KPIs, trends and slicers so a manager can answer "which store, which category, which month?" in seconds.
5. **End with decisions, not charts** — every business question in §1.5 is answered in `insights/Insights_and_Recommendations.md` with a recommendation and expected impact.

1.5 Business questions (the backbone of the project)

Profitability

- BQ1. What are total Sales, Profit and Profit Margin — overall, by year and by month?
- BQ2. Which categories, sub-categories and products are most and least profitable? Which products are loss-making or near-zero margin?

Discounting

- BQ3. Does a higher discount increase sales volume enough to justify the margin loss? At what discount level does a sale stop being profitable?

Seasonality

- BQ4. Where is seasonality strongest, and why is festive-season margin lower?

Stores & geography

- BQ5. Which stores, cities and states perform best on Sales *and* Margin? Which high-sales stores are quietly low-margin?

Customers

- BQ6. How do Regular / Premium / VIP / New segments differ in sales and margin? How strong is repeat purchasing?

Operations

- BQ7. What is the split of Order Status and Payment Method, and what sales value is stuck in Cancelled / Returned orders?
- BQ8. What is the return rate by category, product and reason — and how much of it is preventable (damaged, wrong item, duplicate)?

1.6 KPIs defined for this project

KPI	Formula
Total Sales	SUM(Quantity × UnitPrice × (1 − Discount))
Total Profit	SUM(Sales − Cost)
Profit Margin %	Total Profit ÷ Total Sales
Average Order Value	Total Sales ÷ Distinct Orders (5,905 orders → ₹19,250 per order)
Loss-making Lines %	Lines with Profit < 0 ÷ Total Lines (14.8%)
Return Rate	Returns ÷ Order Lines in that category
Repeat Customer Rate	Customers with ≥ 2 distinct orders ÷ Customers who ordered

1.7 Scope & stakeholders

- **In scope:** 2024–2025 transactions, 20 stores, 1,000 customers, 200 products, 620 returns

- **Out of scope:** inventory, marketing spend, and employee data (not in the dataset — stated honestly, not assumed)
- **Stakeholders:** Store managers (store ranking), Category managers (product / discount decisions), Operations (returns), Leadership (overall margin and festive strategy)

1.8 Tools — and why each one

Tool	Role in this project
Excel	Cleaning, Power Query, PivotTables, KPI workbook — the tool managers already use
SQL (MySQL)	Structured, reproducible answers to BQ1–BQ8; joins, CTEs, window functions
Python (Pandas, Matplotlib)	EDA, correlation, discount-band analysis, chart evidence
Power BI	Interactive dashboard: KPI cards, trend, category / store pages, DAX measures
GitHub	Full project story: problem → data → analysis → insights → recommendations